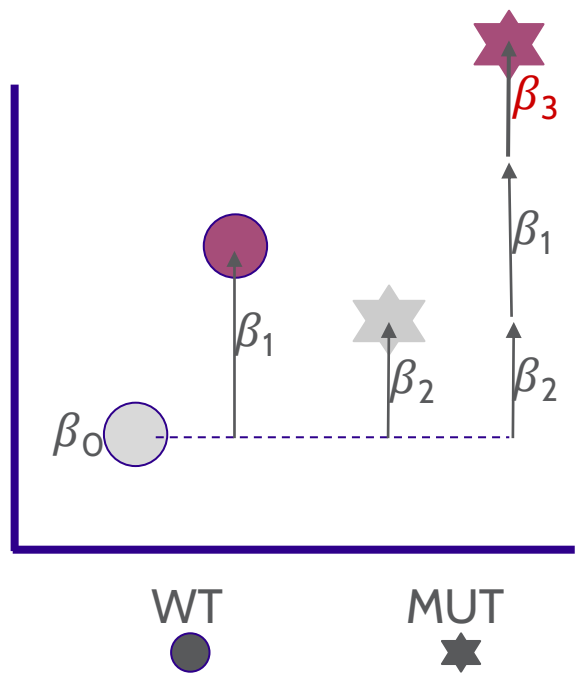


MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

	drug	genotype
sample1	Pink	WT
sample2	Pink	WT
sample3	Pink	MUT
sample4	Pink	MUT
sample5	White	WT
sample6	White	WT
sample7	White	MUT
sample8	White	MUT

Design:
~ drug + genotype + drug:genotype



Null hypothesis:

Pink vs White (WT)

$$\beta_1 = 0$$

Pink vs White (MUT)

$$\beta_1 + \beta_3 = 0$$

WT vs MUT (White)

$$\beta_2 = 0$$

WT vs MUT (Pink)

$$\beta_2 + \beta_3 = 0$$

Interaction ("Difference of differences"):

$$\beta_3 = 0$$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

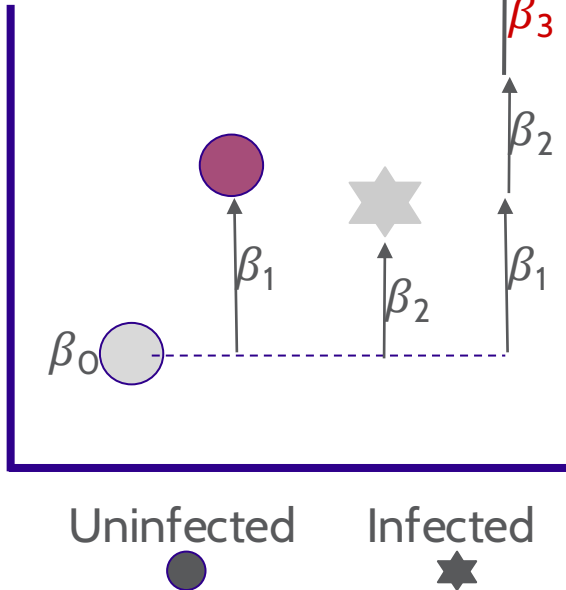
SampleName	Status	TimePoint
SRR7657878	Infected	d11
SRR7657881	Infected	d11
SRR7657880	Infected	d11
SRR7657874	Infected	d33
SRR7657882	Uninfected	d33
SRR7657872	Infected	d33
SRR7657877	Uninfected	d11
SRR7657876	Uninfected	d11
SRR7657879	Uninfected	d11
SRR7657883	Uninfected	d33
SRR7657873	Infected	d33
SRR7657875	Uninfected	d33

Design: ~ TimePoint + Status + TimePoint:Status

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: ~ TimePoint + Status + TimePoint:Status

● - d11
● - d33



Model:

$$Y = \beta_0 + \beta_1 T_{d33} + \beta_2 S_{Inf} + \beta_3 T_{d33} S_{Inf} + \epsilon$$

$\beta_0 \rightarrow$ Expression in uninfected mice at day 11

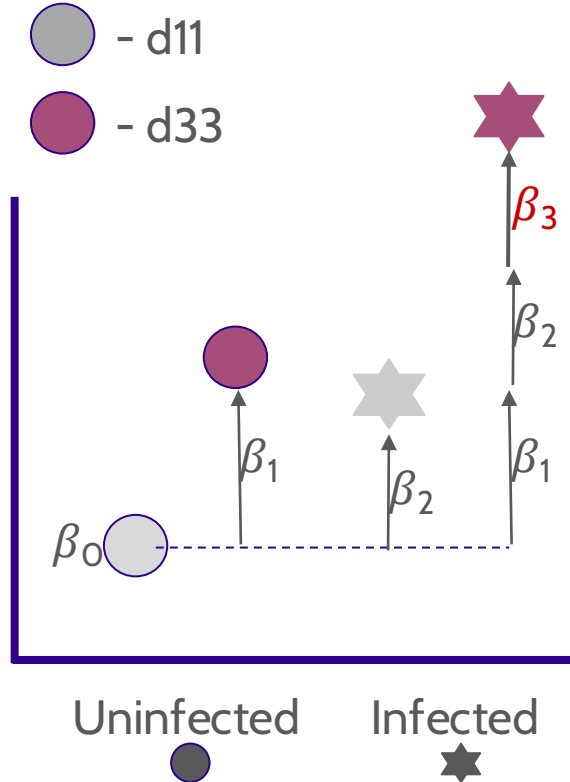
$\beta_1 \rightarrow$ Effect of moving to day 33

$\beta_2 \rightarrow$ Effect of being infected

$\beta_3 \rightarrow$ Additional change due to interaction of time and infection

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Coefficient names in DESeq2

β_0 : *Intercept*

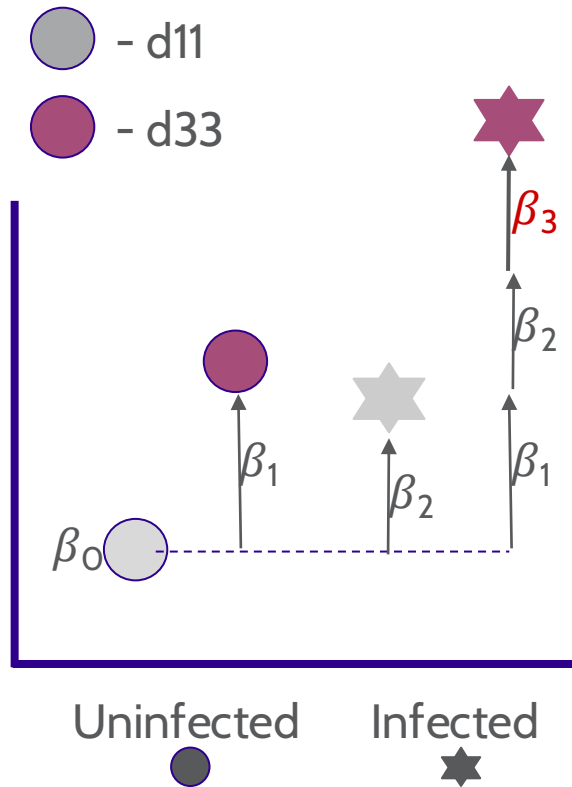
β_1 : *TimePoint_d33_vs_d11*

β_2 : *Status_Infected_vs_Uninfected*

β_3 : *TimePointd33.StatusInfected*

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: ~ TimePoint + Status + TimePoint:Status



Coefficient names in DESeq2

β_0 : *Intercept*

β_1 : *TimePoint_d33_vs_d11*

β_2 : *Status_Infected_vs_Uninfected*

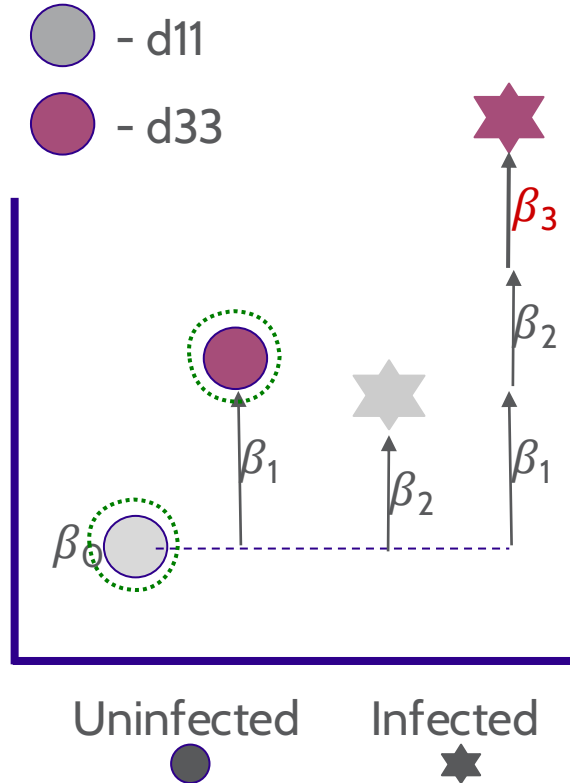
β_3 : *TimePointd33.StatusInfected*

```
resultsNames(ddsObj.interaction)
```

```
## [1] "Intercept" "TimePoint_d33_vs_d11"  
## [3] "Status_Infected_vs_Uninfected" "TimePointd33.StatusInfected"
```

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: d33 vs d11 **when** Uninfected

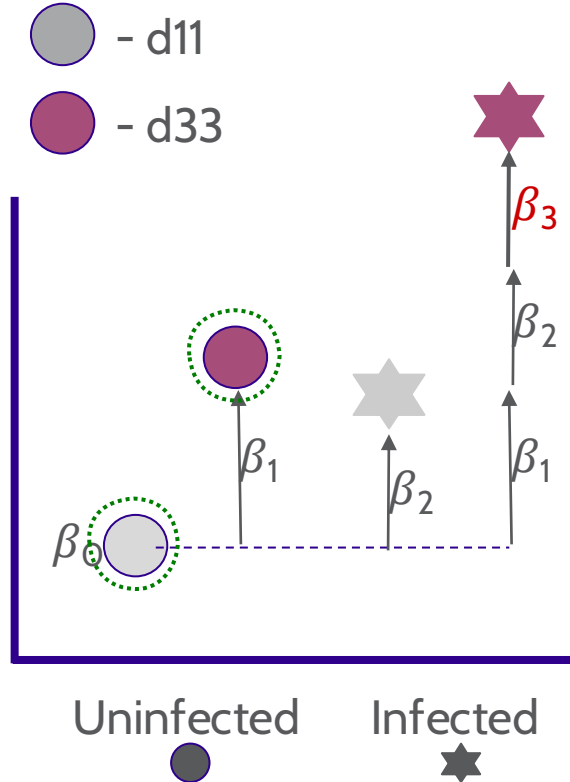
Gene Expression:

Uninfected d11 = β_0

Uninfected d33 = $\beta_0 + \beta_1$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: d33 vs d11 **when** Uninfected

Gene Expression:

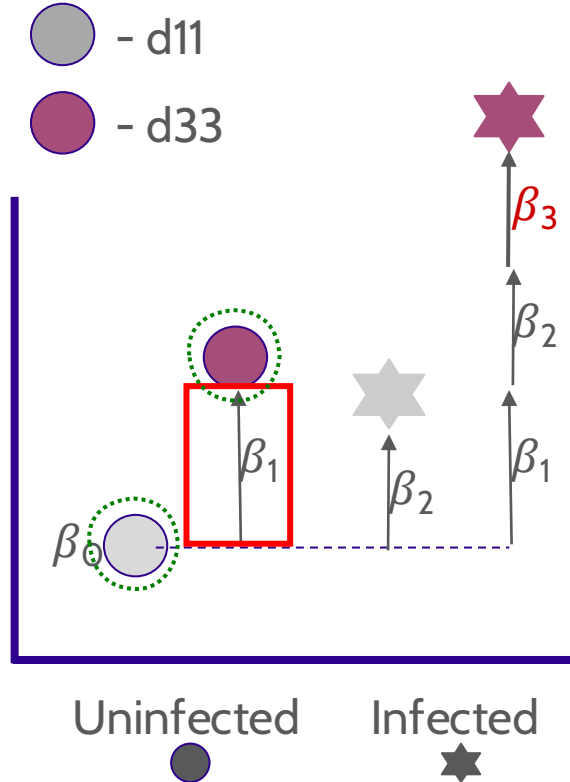
Uninfected d11 = β_0

Uninfected d33 = $\beta_0 + \beta_1$

d33 vs d11 **when** Uninfected = $(\beta_0 + \beta_1) - \beta_0 = \beta_1$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: d33 vs d11 **when** Uninfected

Gene Expression:

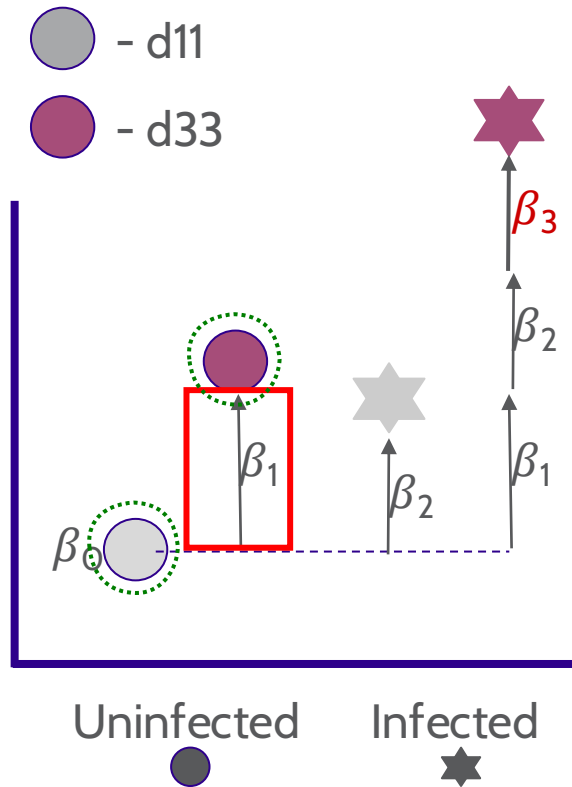
Uninfected d11 = β_0

Uninfected d33 = $\beta_0 + \beta_1$

d33 vs d11 **when** Uninfected = $(\beta_0 + \beta_1) - \beta_0 = \beta_1$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: d33 vs d11 **when** Uninfected

Gene Expression:

Uninfected d11 = β_0

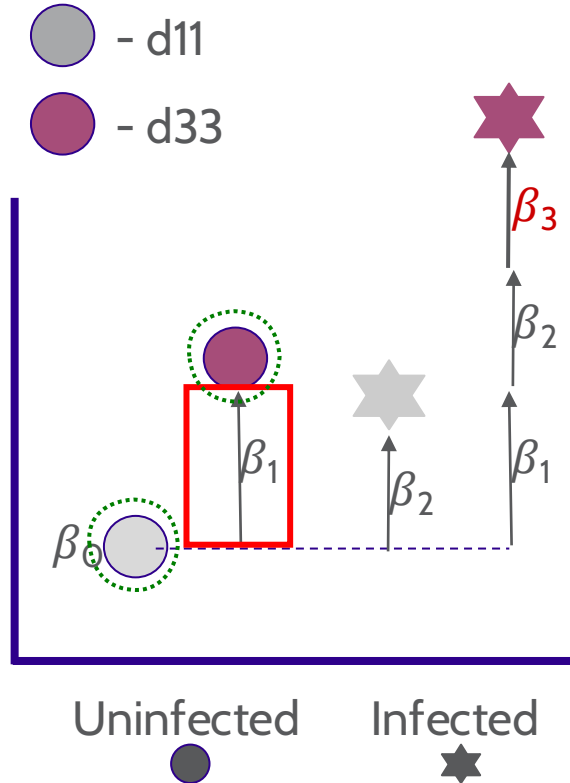
Uninfected d33 = $\beta_0 + \beta_1$

d33 vs d11 **when** Uninfected = $(\beta_0 + \beta_1) - \beta_0 = \beta_1$

β_1 : *TimePoint_d33_vs_d11*

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: d33 vs d11 **when Uninfected**

Gene Expression:

Uninfected d11 = β_0

Uninfected d33 = $\beta_0 + \beta_1$

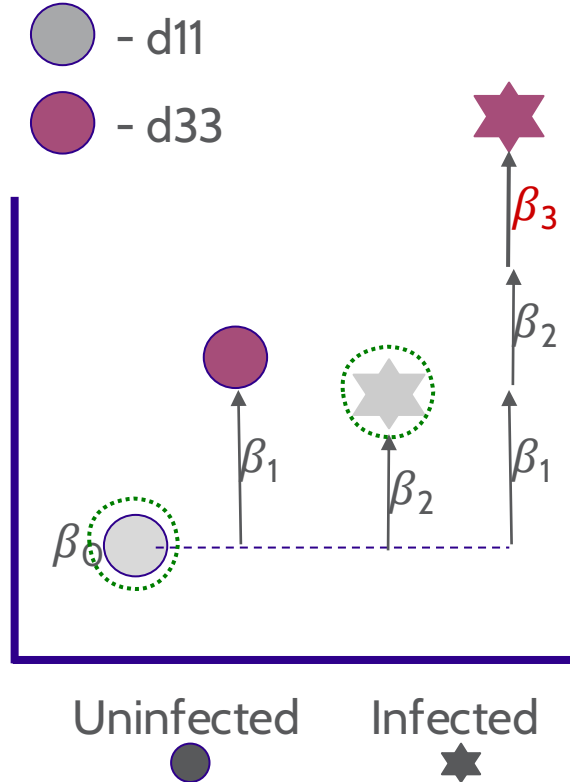
d33 vs d11 **when Uninfected** = $(\beta_0 + \beta_1) - \beta_0 = \beta_1$

β_1 : *TimePoint_d33_vs_d11*

```
res <- results(ddsObj, name = "TimePoint_d33_vs_d11")
```

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: Infected vs Uninfected at d11

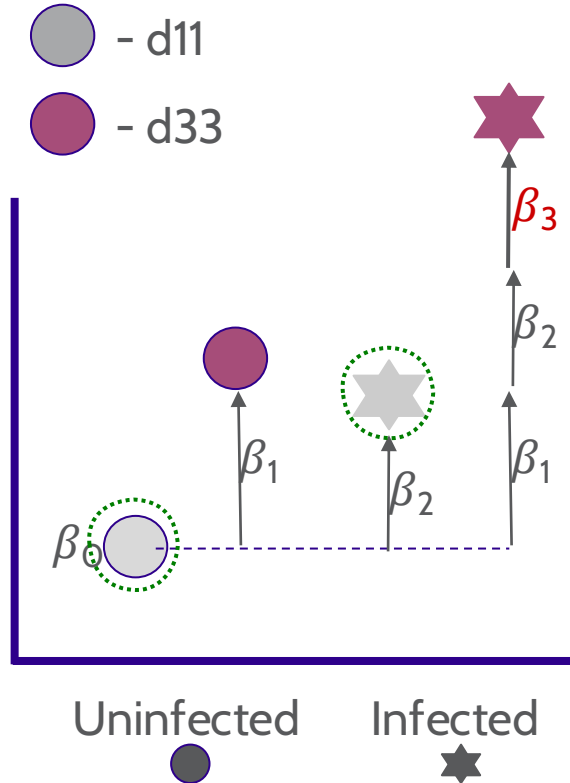
Gene Expression:

Uninfected d11 = β_0

Infected d11 = $\beta_0 + \beta_2$

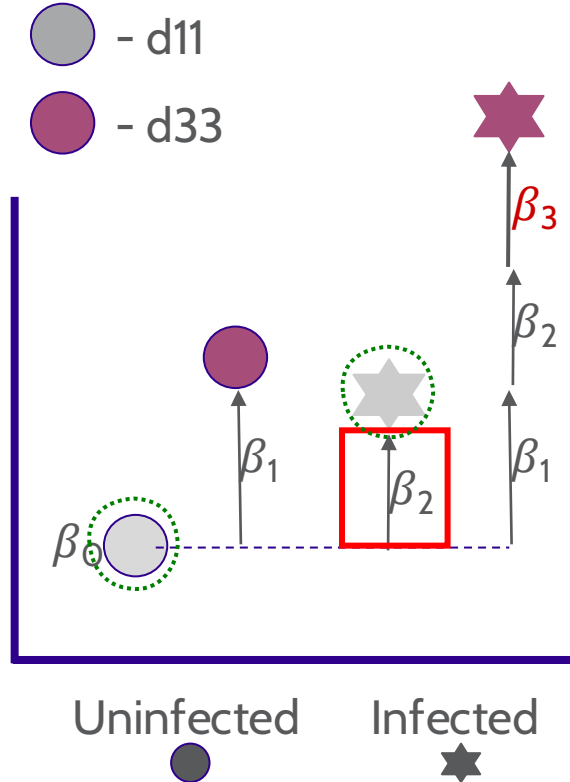
MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: Infected vs Uninfected at d11

Gene Expression:

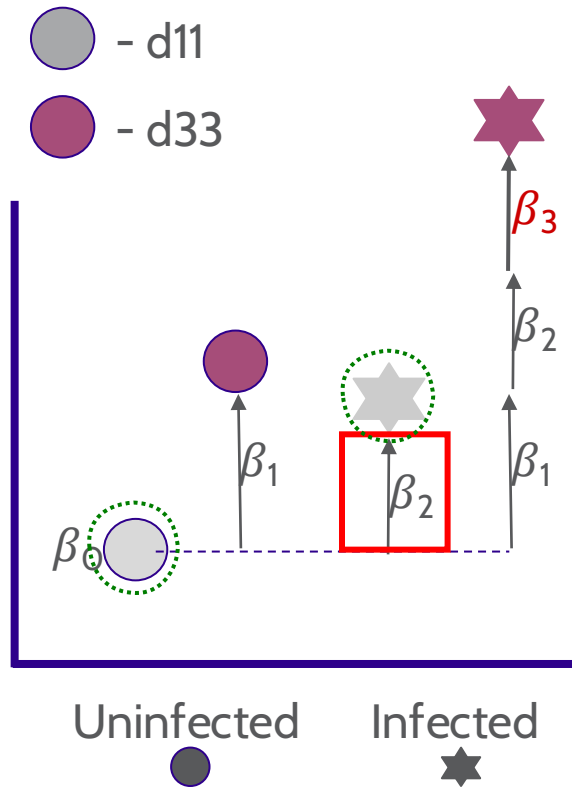
Uninfected d11 = β_0

Infected d11 = $\beta_0 + \beta_2$

Infected vs Uninfected = $(\beta_0 + \beta_2) - \beta_0 = \beta_2$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: Infected vs Uninfected at d11

Gene Expression:

Uninfected d11 = β_0

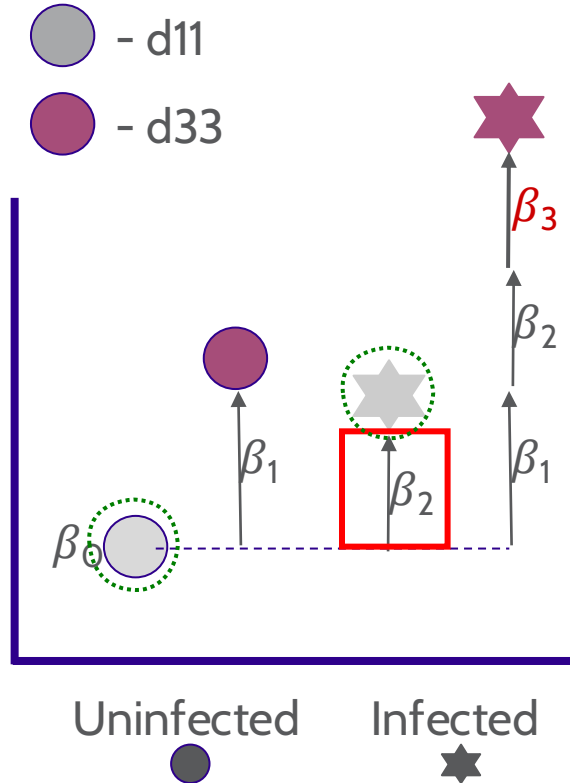
Infected d11 = $\beta_0 + \beta_2$

Infected vs Uninfected = $(\beta_0 + \beta_2) - \beta_0 = \beta_2$

β_2 : *Status_Infected_vs_Uninfected*

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



Contrast: Infected vs Uninfected at d11

Gene Expression:

Uninfected d11 = β_0

Infected d11 = $\beta_0 + \beta_2$

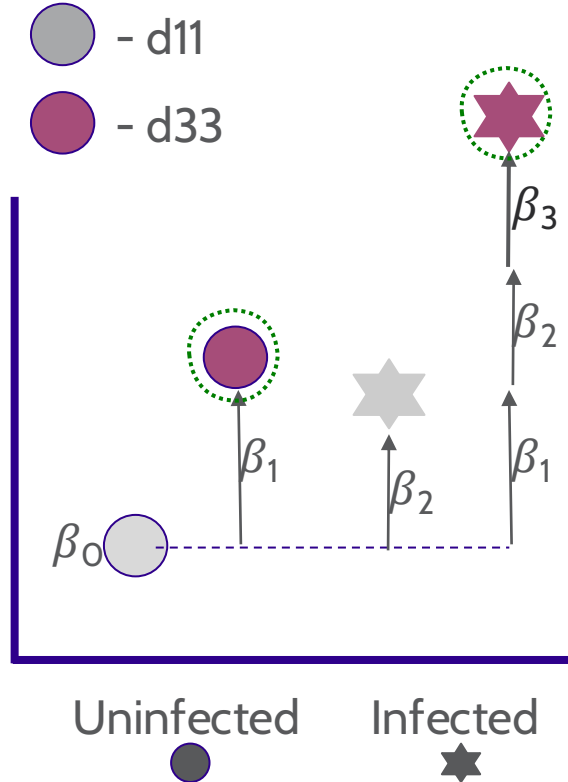
Infected vs Uninfected = $(\beta_0 + \beta_2) - \beta_0 = \beta_2$

β_2 : *Status_Infected_vs_Uninfected*

```
res <- results(ddsObj, name = "Status_Infected_vs_Uninfected")
```

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: Infected vs Uninfected at d33

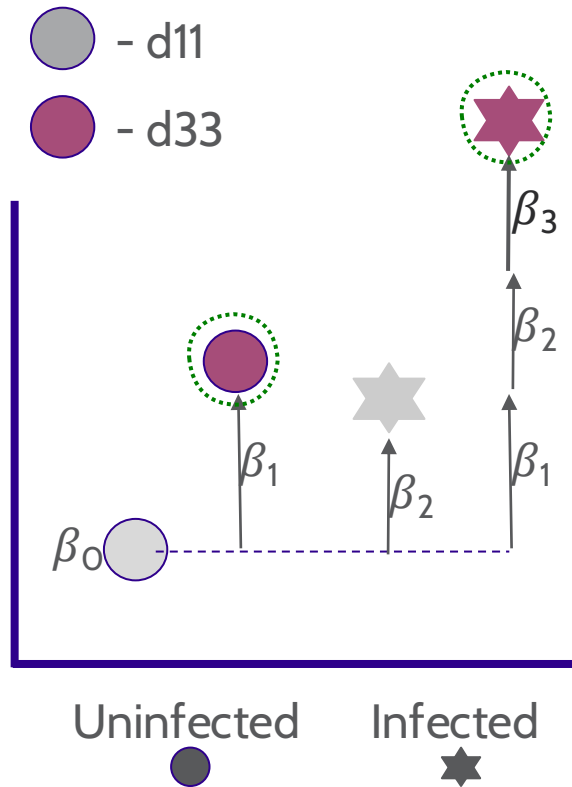
Gene Expression:

$$\text{Uninfected d33} = \beta_0 + \beta_1$$

$$\text{Infected d33} = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: Infected vs Uninfected at d33

Gene Expression:

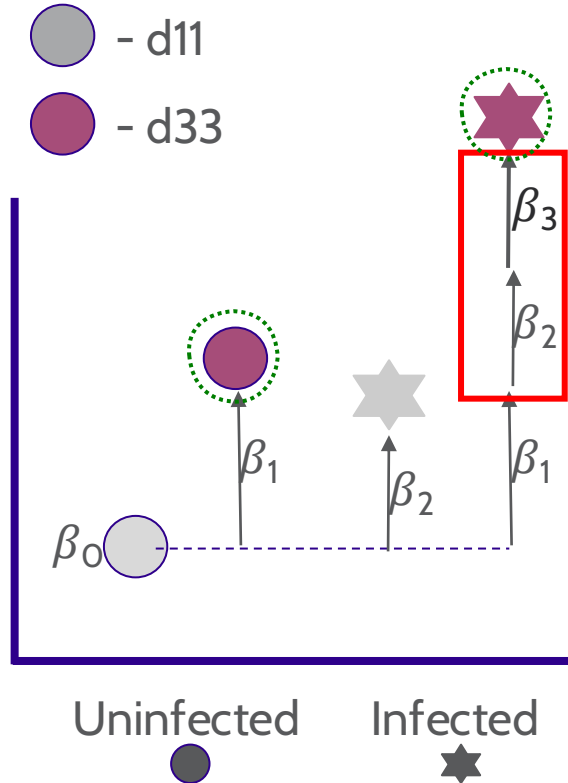
$$\text{Uninfected d33} = \beta_0 + \beta_1$$

$$\text{Infected d33} = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

$$\text{Infected vs Uninfected} = (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_1) = \beta_2 + \beta_3$$

MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: $\sim \text{TimePoint} + \text{Status} + \text{TimePoint}:\text{Status}$



Contrast: Infected vs Uninfected at d33

Gene Expression:

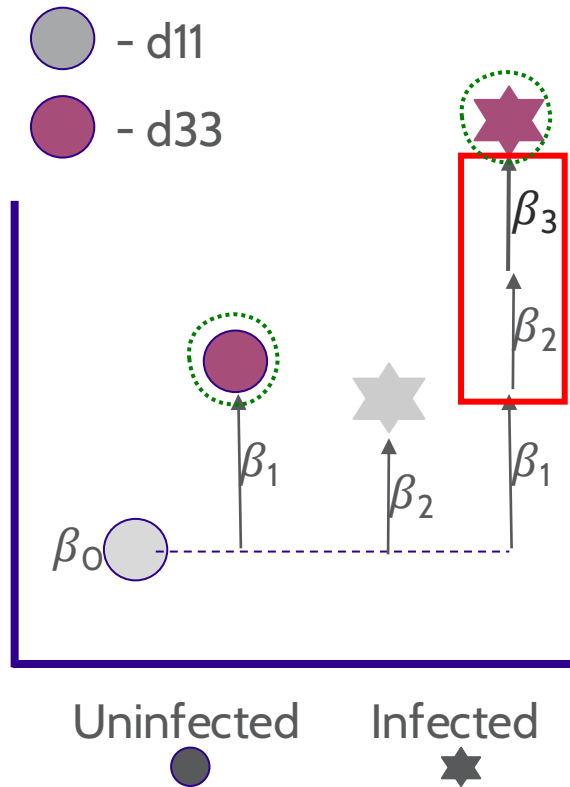
$$\text{Uninfected d33} = \beta_0 + \beta_1$$

$$\text{Infected d33} = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

$$\text{Infected vs Uninfected} = (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_1) = \beta_2 + \beta_3$$

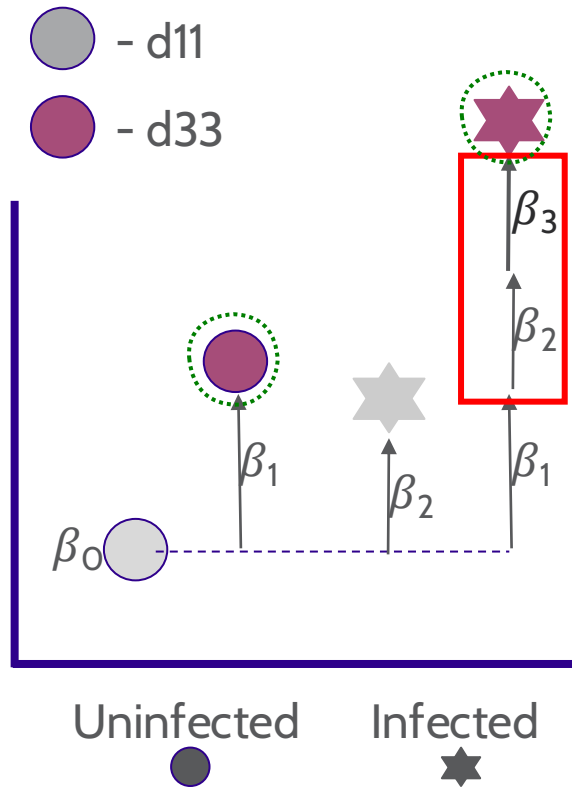
MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: `~ TimePoint + Status + TimePoint:Status`



MODEL DESIGNS | TWO FACTORS - INTERACTION MODEL

Design: ~ TimePoint + Status + TimePoint:Status



Contrast: Infected vs Uninfected at d33

Gene Expression:

$$\text{Uninfected d33} = \beta_0 + \beta_1$$

Infected d33 = $\beta_0 + \beta_1 + \beta_2 + \beta_3$

$$\text{Infected vs Uninfected} = (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_1) = \beta_2 + \beta_3$$

$$\beta_2: \text{Status_Infected_vs_Uninfected}$$
$$\beta_3: \text{TimePointd33.StatusInfected}$$
[illegible]